# Лабораторная работа

Студент: Решетников Артём Денисович, М8О-101БВ-26

## Цель работы:

Мой ник на *GitHub*: **artyom_reshetnikov/mai**. Мой ник на *Docker Hub*: **artyomreshetnikov**.

# Этап 1. Свой образ и Docker Hub

## 1.1. Сборка образа

```
artemreshetnikov@MacBook-Pro--Artyom frontend % docker build -t artyomreshetnikov/lab-frontend:1.0 .
[+] Building 0.9s (8/8) FINISHED                                                docker:desktop-linux
 => [internal] load build definition from Dockerfile                                             0.0s
 => => transferring dockerfile: 213B                                                             0.0s
 => [internal] load metadata for docker.io/library/nginx:alpine                                  0.8s
 => [internal] load .dockerignore                                                                0.0s
 => => transferring context: 2B                                                                  0.0s
 => [1/3] FROM docker.io/library/nginx:alpine@sha256:a3554c0d0b48177b1413c26e01f695334428020d9   0.0s
 => => resolve docker.io/library/nginx:alpine@sha256:a3554c0d0b48177b1413c26e01f695334428020d9   0.0s
 => [internal] load build context                                                                0.0s
 => => transferring context: 64B                                                                 0.0s
 => CACHED [2/3] COPY index.html /usr/share/nginx/html/index.html                                0.0s
 => CACHED [3/3] COPY default.conf /etc/nginx/conf.d/default.conf                                0.0s
 => exporting to image                                                                           0.0s
 => => exporting layers                                                                          0.0s
 => => exporting manifest sha256:1f50fdfe440ea29902a0dc3e283c593bc35bb38a395795f8986f70c23d46b   0.0s
 => => exporting config sha256:0737d143dc1bb20b67f9a528648a668b4be19d1663aa31bc15d99891af3fa95   0.0s
 => => exporting attestation manifest sha256:2b759d7e7b93fb561340d26a457134e543d9b159abd3b5096   0.0s
 => => exporting manifest list sha256:c13fe278406cc74885de89b4a4d7600a003f21afad1ee3acd8b255b5   0.0s
 => => naming to docker.io/artyomreshetnikov/lab-frontend:1.0                                    0.0s
 => => unpacking to docker.io/artyomreshetnikov/lab-frontend:1.0                                 0.0s
artemreshetnikov@MacBook-Pro--Artyom frontend %
```

## 1.2. Запуск контейнера и страница в браузере

```
artemreshetnikov@MacBook-Pro--Artyom frontend % docker run -d -p 8080:80 artyomreshetnikov/lab-fronte
nd:1.0
4f3001bf22f818ad0c927a20418106ed384c00b3e8654c05860cdf33f1bd9cd0
```




# Решетников Артём Денисович

Github: artyom_reshetnikov/mai

Docker Hub: artyomreshetnikov

Вызвать API

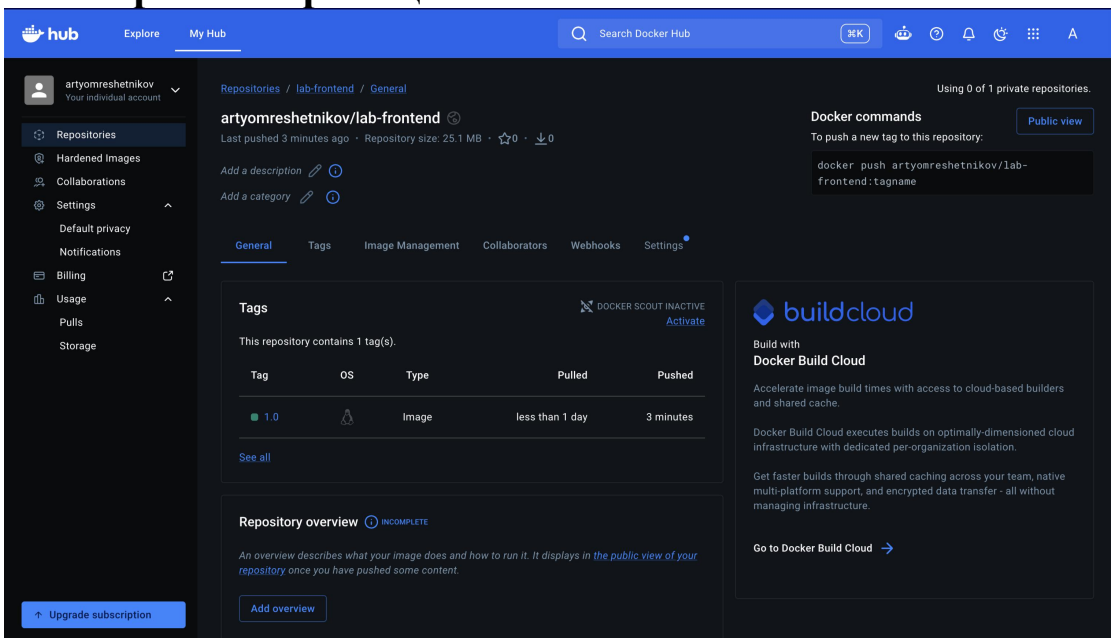
# 1.3. Отправка образа в Docker Hub

```
artemreshetnikov@MacBook-Pro--Artyom frontend % docker login
Authenticating with existing credentials... [Username: artyomres
hetnikov]

i Info → To login with a different account, run 'docker logout'
followed by 'docker login'


Login Succeeded
artemreshetnikov@MacBook-Pro--Artyom frontend % docker push arty
omreshetnikov/lab-frontend:1.0
The push refers to repository [docker.io/artyomreshetnikov/lab-f
rontend]
1408044c7403: Pushed
3fc421ad9e66: Pushed
d3531b7a4f21: Pushed
44136fa355b3: Pushed
7b9d02243238: Pushed
f523b340a6c0: Pushed
dbb1ef19121a: Pushed
567a969e0221: Pushed
c2fefbef5314: Pushed
cbe54b031f3e: Pushed
3ebb0e1c1245: Pushed
e36251012972: Pushed
1.0: digest: sha256:0c4476c2e3a28185448514d4c267e4cb759288bd0b7b
c0fdf7e567bfa8862c78 size: 856
artemreshetnikov@MacBook-Pro--Artyom frontend %
```

# 1.4. Образ на странице Docker Hub



### 1.5. Удаление образа и скачивание обратно

```
[docker pull artyomreshetnikov/lab-frontend:1.0                          ]
Untagged: artyomreshetnikov/lab-frontend:1.0
Deleted: sha256:0c4476c2e3a28185448514d4c267e4cb759288bd0b7bc0fdf7e567
bfa8862c78
1.0: Pulling from artyomreshetnikov/lab-frontend
e36251012972: Pull complete
1408044c7403: Pull complete
d3531b7a4f21: Pull complete
cbe54b031f3e: Pull complete
c2fefbef5314: Pull complete
f523b340a6c0: Pull complete
dbb1ef19121a: Pull complete
7b9d02243238: Pull complete
3ebb0e1c1245: Pull complete
567a969e0221: Pull complete
44136fa355b3: Download complete
3fc421ad9e66: Download complete
Digest: sha256:0c4476c2e3a28185448514d4c267e4cb759288bd0b7bc0fdf7e567b
fa8862c78
Status: Downloaded newer image for artyomreshetnikov/lab-frontend:1.0
docker.io/artyomreshetnikov/lab-frontend:1.0
```

# Этап 2. Docker Compose с сетью и директориями

### 2.1. Файл *docker-compose.yml*

```
[artemreshetnikov@MacBook-Pro--Artyom docker-lab % cat docker-compose.y]
ml
services:
  frontend:
    image: artyomreshetnikov/lab-frontend:1.0
    ports:
      - "8080:80"
    depends_on: # порядок запуска
      - api
    networks:
      - appnet # добавление в общую сеть
  api:
    image: contanious/whoami
    networks:
      - appnet # аналогично
  logger:
    image: busybox
    command: sh -c 'while true; do date >> /data/log.txt; sleep 2; don
e'
    # логгирование в файл логов каждые 2 сек
    volumes:
      - loggerdata:/data # подключение директории для логов
    networks:
        - appnet # аналогично
  volumes:
    loggerdata:

  networks:
    appnet:%
```

## 2.2. Запуск всех сервисов

```
artemreshetnikov@MacBook-Pro--Artyom docker-lab % docker compose up -d

[+] up 5/5
 ✔ Network docker-lab_appnet       Created                          0.0s
 ✔ Container docker-lab-api-1      Started                         30.3s
 ✔ Container docker-lab-logger-1   Started                         30.2s
 ! api                             The requested image's plat...    0.0s
 ✔ Container docker-lab-frontend-1 Started                         30.3s
artemreshetnikov@MacBook-Pro--Artyom docker-lab % docker compose ps
NAME                   IMAGE                                COMMAND
                SERVICE    CREATED          STATUS          PORTS
docker-lab-api-1       containous/whoami                    "/whoami"
                api        43 seconds ago   Up 12 seconds   80/tcp
docker-lab-frontend-1  artyomreshetnikov/lab-frontend:1.0   "/docker-
entrypoint.…"   frontend   43 seconds ago   Up 12 seconds   0.0.0.0:80
80->80/tcp, [::]:8080->80/tcp
docker-lab-logger-1    busybox                              "sh -c 'w
hile true; …"   logger     43 seconds ago   Up 42 seconds
artemreshetnikov@MacBook-Pro--Artyom docker-lab %
```

## 2.3. Проверка на то, что контейнеры находят друг друга по имени

```
artemreshetnikov@MacBook-Pro--Artyom docker-lab % docker compose exec
frontend sh
/ # curl http://api
Hostname: c23d624aa83c
IP: 127.0.0.1
IP: ::1
IP: 172.18.0.3
RemoteAddr: 172.18.0.4:37572
GET / HTTP/1.1
Host: api
User-Agent: curl/8.22.0
Accept: */*

/ # exit
```

## 2.4. Том для logger

```
artemreshetnikov@MacBook-Pro--Artyom docker-lab % docker volume ls | g
rep logger
local     docker-lab_loggerdata
artemreshetnikov@MacBook-Pro--Artyom docker-lab %
```

## 2.5. Строки, которые пишет logger

```
artemreshetnikov@MacBook-Pro--Artyom docker-lab % docker compose exec
logger cat /data/log.txt
Sat Oct 24 10:53:18 UTC 2026
Sat Oct 24 10:53:20 UTC 2026
Sat Oct 24 10:53:22 UTC 2026
Sat Oct 24 10:53:24 UTC 2026
Sat Oct 24 10:53:26 UTC 2026
```

(На данном этапе в log.txt 474 строки)

```
artemreshetnikov@MacBook-Pro--Artyom docker-lab % docker compose exec
logger wc -l /data/log.txt
474 /data/log.txt
```

# Этап 3. Проверка сохранения данных при пересозданиии контейнеров

## 3.1. Удаление контейнеров без удаления тома и повторный запуск

```
artemreshetnikov@MacBook-Pro--Artyom docker-lab % docker
compose down
[+] down 2/3
 ⠸ Container docker-lab-logger-
1   Stopping                    1.7s
[+] down 4/4
 ✔ Container docker-lab-logger-
1   Removed                     3.1s
 ✔ Network docker-
lab_appnet       Removed                  0.1s
```

```
artemreshetnikov@MacBook-Pro--Artyom docker-lab % docker compose up -d

[+] up 5/5
 ✔ Network docker-lab_appnet       Created                      0.0s
 ✔ Container docker-lab-logger-1   Started                      0.3s
 ✔ Container docker-lab-api-1      Started                      0.2s
 ! api                             The requested image's platf... 0.0s
 ✔ Container docker-lab-frontend-1 Started                      0.2s
artemreshetnikov@MacBook-Pro--Artyom docker-lab %
```

## 3.2. Файл логгера после пересоздания

```
artemreshetnikov@MacBook-Pro--Artyom docker-lab % docker compose exec
logger cat /data/log.txt
Sat Oct 24 10:53:18 UTC 2026
Sat Oct 24 10:53:20 UTC 2026
Sat Oct 24 10:53:22 UTC 2026
Sat Oct 24 10:53:24 UTC 2026
```

```
artemreshetnikov@MacBook-Pro--Artyom docker-lab % docker compose exec
logger wc -l /data/log.txt
710 /data/log.txt
artemreshetnikov@MacBook-Pro--Artyom docker-lab %
```

(старые записи остались и дозаписались новые)

## 3.3. Удаление тома флагом *-v* + *проверка удаления данных*

```
artemreshetnikov@MacBook-Pro--Artyom docker-lab % docker compose down -v
[+] down 5/5
 ✔ Container docker-lab-logger-1   Removed                          3.1s
 ✔ Container docker-lab-frontend-1 Removed                          0.2s
 ✔ Container docker-lab-api-1      Removed                          0.1s
 ✔ Volume docker-lab_loggerdata    Removed                          0.0s
 ✔ Network docker-lab_appnet       Removed                          0.1s
artemreshetnikov@MacBook-Pro--Artyom docker-lab % docker compose up -d

[+] up 6/6
 ✔ Network docker-lab_appnet       Created                          0.0s
 ✔ Volume docker-lab_loggerdata    Created                          0.0s
 ✔ Container docker-lab-logger-1   Started                          0.3s
 ✔ Container docker-lab-api-1      Started                          0.2s
 ! api                             The requested image's platf...   0.0s
 ✔ Container docker-lab-frontend-1 Started                          0.2s
artemreshetnikov@MacBook-Pro--Artyom docker-lab % docker compose exec logger sh -c "cat /data/log.txt"
Sat Oct 24 11:21:59 UTC 2026
Sat Oct 24 11:22:01 UTC 2026
Sat Oct 24 11:22:03 UTC 2026
Sat Oct 24 11:22:05 UTC 2026
Sat Oct 24 11:22:07 UTC 2026
Sat Oct 24 11:22:09 UTC 2026
Sat Oct 24 11:22:11 UTC 2026
Sat Oct 24 11:22:13 UTC 2026
Sat Oct 24 11:22:15 UTC 2026
artemreshetnikov@MacBook-Pro--Artyom docker-lab %
```